Analysis request

The prompt

This report was produced by Claude, working through the PACER 3.2.0 tools, in response to the request below, grouped into the steps it asks for.

GOAL	I want to complete a comprehensive exploratory analysis of these data `C:\PACER\IrtEstimation\data\gdina_responses.csv`.
STEP 1 · CTT	I first want you to use PACER and perform a complete CTT analysis. Identify items that should be reviewed or flagged for any reason and explain why.
STEP 2 · IRT	Then, I am not sure if IRT or CDM is the best way to approach the analysis of these items. First, run some IRT analyses and generate ability estimates. Decide if IRT models fit these data and if so which one do you recommend.
STEP 3 · CDM	Then, run G-DINA models, and perhaps other CDMs, in PACER using this Q matrix `C:\PACER\IrtEstimation\data\gdina_qmatrix.csv`. Go ahead and also build skill mastery profiles.
STEP 4 · COMPARE	Compare the IRT ability estimates to the skill profiles.
CONSTRAINTS & OUTPUT	Use only tools available in PACER and build visual displays for me. Summarize your findings and results, show me tables and fit statistics, but build a complete narrative I can review.

The report follows on the next page, unchanged from the interactive version except for print layout. Hover tooltips from the web version are not available in print.

PACER 3.2.0 · exploratory psychometric review · 14 Oct 2026

Is this a single scale or a set of skills?

A classical, IRT, and cognitive-diagnostic look at `gdina_responses.csv`, 10 dichotomous items taken by 1,000 examinees and mapped to three attributes by `gdina_qmatrix.csv`.

1,000 examinees **10** items, 0/1, no missing data **3** attributes (A1–A3)

Models: CTT · 1PL · 2PL · 3PL · DINA · DINO · A-CDM · LLM · R-RUM · G-DINA

Bottom line

1. **Use G-DINA as the primary model, with the LLM as its parsimonious alternative.** Diagnostic models fit far better than IRT. Among the six CDMs, G-DINA has the best AIC (12,219), and likelihood-ratio tests reject every reduced model against it. R-RUM edges it on BIC by 4 points (12,416 vs. 12,420), but R-RUM gives a degenerate solution (below). The LLM reproduces G-DINA's profiles for 91% of examinees with 8 fewer parameters.
2. **No single reduced model suits every item.** V5 behaves conjunctively ("AND"), V6 disjunctively ("OR"), and V4 and V10 additively. A one-rule-for-all-items model always misfits somewhere, which is why the saturated G-DINA wins.
3. **Attribute A1 is weakly identified, and the model you choose decides what it means.** Its two single-skill items, V1 and V8, correlate only .28. G-DINA and LLM build A1 mostly from V1. A-CDM and R-RUM build it almost entirely from V8 (every R-RUM A1 master answered V8 correctly), which cuts A1 mastery from 58% to 30% and empties the 101 profile.
4. **The 2PL is the best IRT option, and it is not adequate on its own.** It clearly beats the 1PL, while the 3PL overfits. Five of 10 items still misfit, which is what you'd expect when a unidimensional model is forced onto three distinct skills.
5. **θ is a blurred summary of the skill profile.** The 2PL ability correlates 0.92 with total expected skill mastery, and profile membership explains 80% of θ variance. But examinees with the same raw score of 5 hold seven different skill profiles. θ weights A3 heavily (r = .76) and A1 lightly (r = .45).
6. **Review V7, V8, V9, and the A1 items first.** V7 is weak under every model. Adding A1 lowers V9 accuracy in G-DINA, A-CDM, LLM and R-RUM alike, which points to a Q-matrix error. V1 and V8 need a content review of what A1 is meant to be.

.668 Cronbach's α (10 items)	**−237** ΔBIC, G-DINA vs. 2PL	**91%** profile agreement, LLM vs. G-DINA	**.88–.93** expected attribute-level classification accuracy (G-DINA)

1 Data and design

The response file is a clean 1,000 × 10 binary matrix with no missing responses and no group column. The Q-matrix assigns each item one to three of three attributes. Five items are single-attribute, four need two attributes, and V9 needs all three. A1 and A3 are each measured by six items, while A2 is measured by only four.

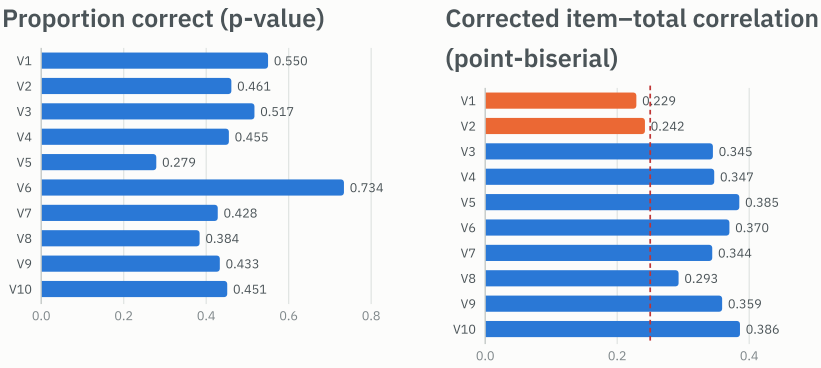
Item	A1	A2	A3	Attributes	Role in the design
V1	1	0	0	1	Single-skill A1 indicator
V2	0	1	0	1	Only single-skill A2 indicator
V3	0	0	1	1	Single-skill A3 indicator
V4	1	0	1	2	A1 + A3 combination
V5	0	1	1	2	A2 + A3 combination
V6	1	1	0	2	A1 + A2 combination
V7	0	0	1	1	Second single-skill A3 indicator
V8	1	0	0	1	Second single-skill A1 indicator
V9	1	1	1	3	Only item requiring all three skills
V10	1	0	1	2	A1 + A3 combination (same as V4)

> A2 rests on only four items, and only one of them (V2) measures A2 alone. Expect A2 classifications to depend heavily on V2, V5, V6, and V9.

2 Classical test theory

PACER's CTT run reports Cronbach's α = **0.668** (SEM = 1.40 raw-score points, SD = 2.44). That is modest for a 10-item test, but typical for a short multi-skill test where items deliberately tap different attributes. Item difficulties span .28 to .73, centered near .45. Every item's α-if-deleted is below the overall .668, so **no single item lowers reliability**. With its default thresholds, PACER flagged no items.

Figure 1. Item difficulty and discrimination



Dashed line on the right panel: a conventional 0.25 review threshold. **V1 (.229) and V2 (.242)** fall below it, and V8 (.293) is marginal. V5 is the hardest item (p = .28) and V6 the easiest (p = .73). Neither is extreme.

Item	p	SD	Pt-bis (adj.)	Biserial	α if deleted	Note
V1	0.550	0.498	0.229	0.288	0.664	▲ Low discrimination
V2	0.461	0.499	0.242	0.304	0.661	▲ Low discrimination
V3	0.517	0.500	0.345	0.433	0.641	✓ OK
V4	0.455	0.498	0.347	0.436	0.640	✓ OK
V5	0.279	0.449	0.385	0.515	0.634	✓ Hardest item
V6	0.734	0.442	0.370	0.499	0.637	✓ Easiest item
V7	0.428	0.495	0.344	0.434	0.641	✓ OK
V8	0.384	0.487	0.293	0.373	0.651	▲ Marginal
V9	0.433	0.496	0.359	0.453	0.638	✓ OK
V10	0.451	0.498	0.386	0.485	0.632	✓ OK

Reading the CTT results in light of the design. The lowest-discriminating items, V1 and V2, are both *single-attribute* items, measuring A1 and A2 respectively. Their low item–total correlations don't necessarily mean they are poor items. They correlate weakly with a total that is dominated by other skills, mostly A3. Section 4 shows both are strong indicators of their own attribute under G-DINA, so CTT alone would have flagged the wrong items here.

3 Item response theory

I calibrated 1PL, 2PL, and 3PL models in PACER (MML-EM, PACER defaults, including the default beta prior on *c* for the 3PL). I then scored examinees with EAP under the 2PL.

Model comparison

Model	−2LL	Params	AIC	BIC	LR test vs. previous	Items misfit
1PL (Rasch-type)	12,578.5	11	12,600.5	12,654.5	—	✕ 10 /
2PL	12,519.5	20	12,559.5	12,657.7	$\chi^2(9) = 59.0$, $p < .001$	▲ 5 /
3PL	12,471.7	30	12,531.7	12,678.9	$\chi^2(10) = 47.8$, $p < .001$*	✕ 9 /

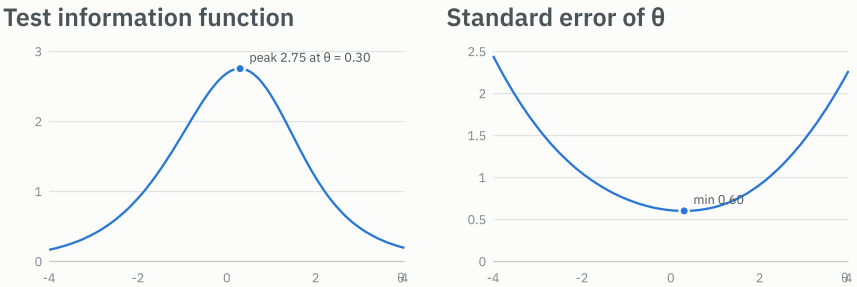
AIC = −2LL + 2k; BIC = −2LL + k·ln(1000). Misfit is PACER's Q1 item-fit flag. *The 3PL is estimated with a prior on *c*, so its likelihood-ratio test is approximate.

Recommendation within IRT: the 2PL. The 1PL's equal-slope assumption clearly fails: the 2PL improves fit by 59 points of −2LL for 9 extra parameters, and every item misfits under the 1PL. The 3PL lowers −2LL further, but BIC penalizes it. It also misfits 9 of 10 items and gives an implausible solution for V3 ($a = 2.73$, $c = .30$). That pattern points to the extra parameters absorbing misfit from multidimensionality, not genuine guessing. BIC is essentially tied between the 1PL and 2PL ($\Delta = 3$). The likelihood-ratio test and item fit both favor the 2PL.

Do IRT models fit these data? Only partially. Under the 2PL, V1, V3, V7, V8, and V9 still misfit. Figure 3 shows why. V7's observed proportion correct rises, then *drops* at mid-to-high θ (0.65 → 0.44), then rises again. A single latent trait cannot produce that pattern. It is what you see when an item depends on one specific skill that doesn't line up with overall ability.

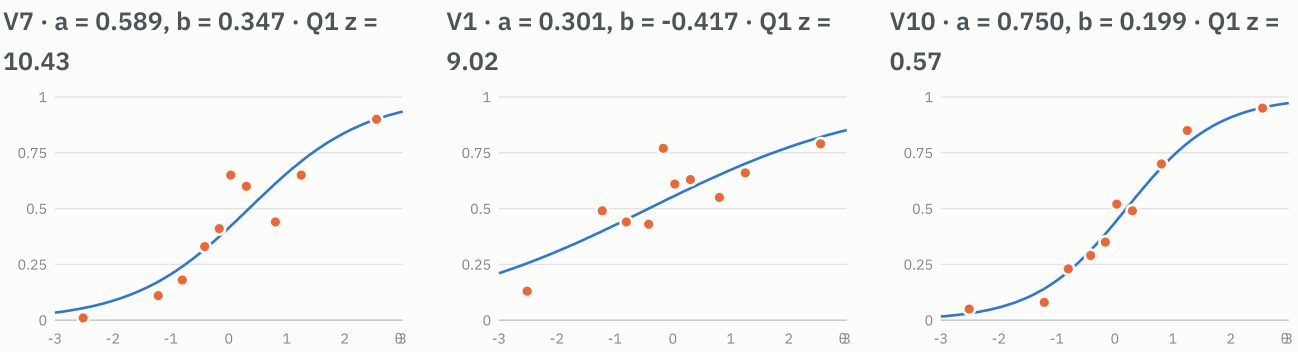
Item	a	b	Q1 z	Fit	3PL a	3PL b	3PL c
V1	0.301	-0.417	9.02	✕ Misfit	0.339	0.035	0.111
V2	0.390	0.258	1.42	✓ OK	0.575	0.539	0.116
V3	0.815	-0.071	6.85	✕ Misfit	2.731	0.561	0.302
V4	0.636	0.204	1.41	✓ OK	0.760	0.637	0.163
V5	0.914	0.865	0.82	✓ OK	1.782	0.908	0.084
V6	0.656	-1.129	1.27	✓ OK	0.931	-0.790	0.103
V7	0.589	0.347	10.43	✕ Misfit	0.597	0.595	0.083
V8	0.394	0.775	5.58	✕ Misfit	0.496	1.001	0.091
V9	0.745	0.276	4.41	✕ Misfit	1.146	0.566	0.147
V10	0.750	0.199	0.57	✓ OK	1.226	0.591	0.186

Figure 2. 2PL test information and conditional standard error



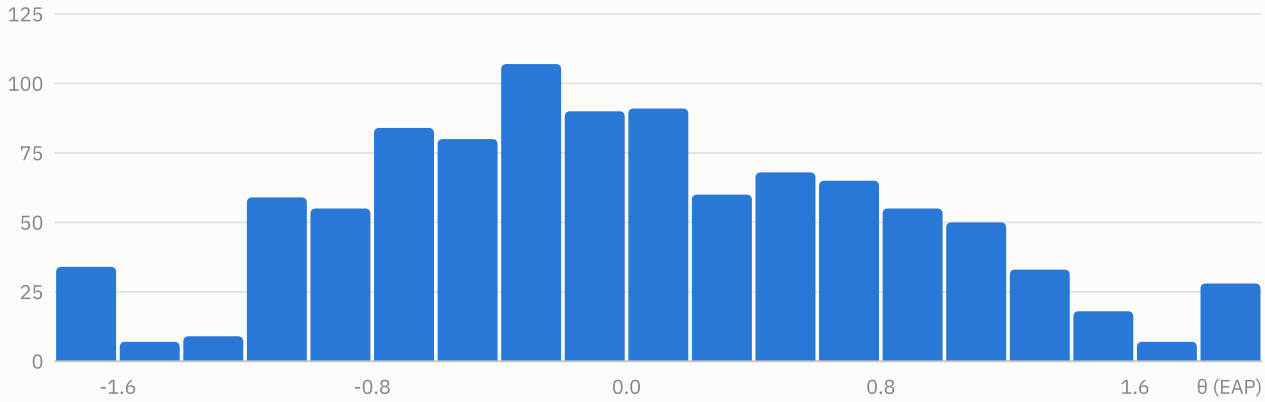
Information peaks at θ = 0.30 (TIF = 2.75, SE = 0.60), which is low. The SE stays above 0.6 everywhere, so θ separates examinees only coarsely. The mean EAP SE is 0.565 on a θ scale with SD 0.82, which implies a marginal reliability of about 0.68, consistent with α.

Figure 3. 2PL item fit: observed vs. model-implied proportion correct



Line: 2PL ICC. Dots: observed proportion correct in 10 θ score groups. **V7** (worst misfit) is non-monotone. **V1** is noisy around a very flat curve (*a* = 0.30). **V10** is shown for contrast as a well-fitting item.

Figure 4. Distribution of 2PL EAP ability estimates



Mean 0.00, SD 0.82, range −1.65 to 1.81. With only 11 possible raw scores and a strong prior, EAP estimates fall on a few dozen distinct values. Peaks at the extremes are the 34 examinees with zero correct and the 28 with all correct.

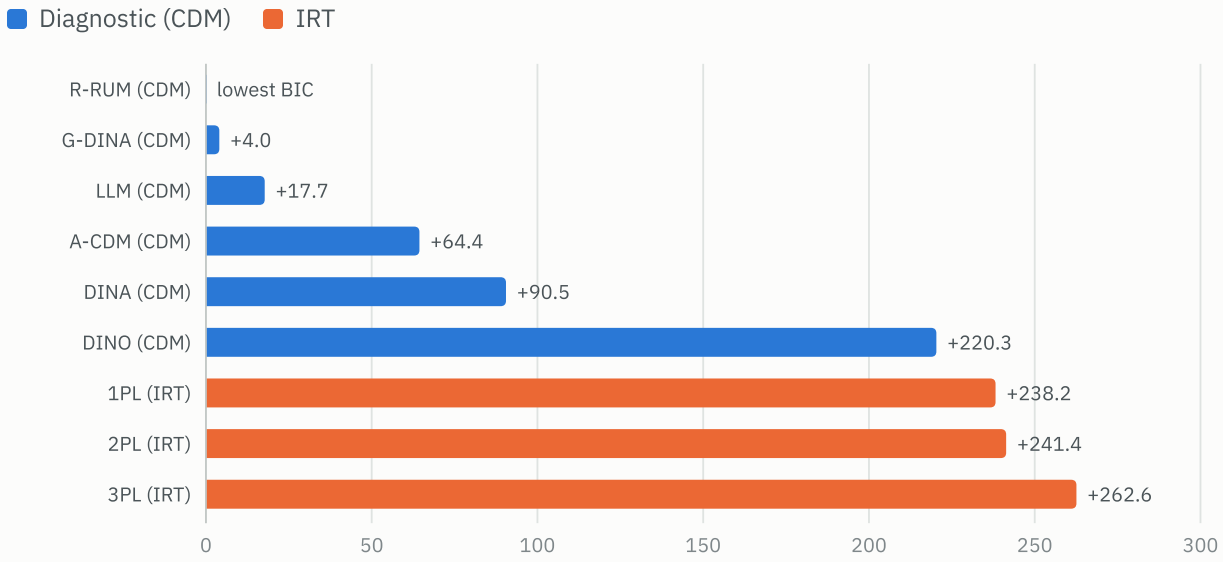
4 Cognitive diagnostic models

I fit six CDMs in PACER with the supplied Q-matrix. They differ in the rule that links the skills an item requires to the chance of answering it correctly.

Model	Rule	Item parameters
DINA	Conjunctive. Success needs *all* required skills; anything less is a guess.	2 per item
DINO	Disjunctive. *Any one* required skill is enough.	2 per item
A-CDM	Additive on the probability scale. Each mastered skill adds a fixed amount to P(correct); no interactions.	k + 1
LLM	Additive on the logit scale (linear logistic model). Skill effects compound like a logistic regression.	k + 1
R-RUM	Additive on the log scale (reduced reparameterized unified model). Each missing skill multiplies P(correct) by a penalty.	k + 1
G-DINA	Saturated. A separate success probability for every combination of required skills, so it nests all five models above.	2^k

k = number of skills the item requires. Every model also estimates 7 latent-class weights for the 8 skill profiles. PACER applies one model to all items in a calibration.

Figure 5. Relative fit of all nine models (BIC, lower is better)



Bars show each model's BIC minus the lowest BIC (R-RUM). R-RUM and G-DINA are within 4 points of each other, and the LLM is 18 points behind. Every CDM except DINO beats every IRT model.

Model	Family	Deviance (−2LL)	Params	AIC	BIC	ΔBIC	LR test vs. G-I
G-DINA	CDM	12,137.15	41	12,219.15	12,420.36	4.0	
R-RUM	CDM	12,188.36	33	12,254.36	12,416.32	0.0	$\chi^2(8) = 51.2$, $p <$
LLM	CDM	12,206.1	33	12,272.1	12,434.05	17.7	$\chi^2(8) = 69.0$, $p <$
A-CDM	CDM	12,252.76	33	12,318.76	12,480.72	64.4	$\chi^2(8) = 115.6$, $p <$
DINA	CDM	12,320.32	27	12,374.32	12,506.83	90.5	$\chi^2(14) = 183.2$, $p <$
DINO	CDM	12,450.14	27	12,504.14	12,636.65	220.3	$\chi^2(14) = 313.0$, $p <$
1PL	IRT	12,578.5	11	12,600.5	12,654.5	238.2	not ne
2PL	IRT	12,519.5	20	12,559.5	12,657.7	241.4	not ne
3PL	IRT	12,471.7	30	12,531.7	12,678.9	262.6	not ne

LR test: deviance difference against G-DINA, with df equal to the difference in parameters. All five reduced CDMs are nested in G-DINA.

Which diagnostic model?

The criteria split, narrowly. AIC ranks G-DINA first (12,219), then R-RUM (12,254) and the LLM (12,272). BIC, which penalizes parameters more heavily, puts R-RUM 4 points ahead of G-DINA, a difference conventionally read as only "positive" evidence. The likelihood-ratio tests are unambiguous: every reduced model fits significantly worse than G-DINA (R-RUM $\chi^2(8) = 51.2$, LLM $\chi^2(8) = 69.0$, A-CDM $\chi^2(8) = 115.6$, all $p < .001$). The 8 extra G-DINA parameters are buying real fit.

G-DINA vs. 2PL: these models are not nested, so I compared them only by information criteria. G-DINA's deviance is 382 points lower for 21 more parameters, and its BIC is 237 points lower, which is decisive evidence on any conventional scale.

Why R-RUM's BIC lead shouldn't decide the matter. R-RUM, and A-CDM with it, reach their fit with a solution that is hard to defend substantively:

- The 101 profile (A1 and A3 without A2) is estimated at exactly 0% in both, while G-DINA and LLM put 13% of examinees there.
- A1 mastery drops to 30% (R-RUM) and 33% (A-CDM), from 59% under G-DINA and 56% under the LLM.
- The reason is that they redefine A1 around V8 instead of V1 (Figure 7 and the A1 table below).

The LLM is the best-behaved reduced model. Its class distribution, mastery rates, and examinee profiles are almost identical to G-DINA's (91% full-profile agreement; 95–98% per attribute). Only V9 violates monotonicity. If a reporting program prefers fewer parameters, the LLM is a defensible choice. G-DINA remains the primary recommendation because it fits significantly better and assumes no condensation rule.

> PACER reports relative fit (AIC/BIC) for CDMs but no absolute fit statistics (e.g., M2, SRMSR) or CDM item-fit tests. "Best of these models" is established. "Fits in an absolute sense" is not.

What the item-level estimates show

Comparing each model's success probabilities with G-DINA's shows which rule each item actually follows. The rules differ across items, which is why no single reduced model fits all of them.

Figure 6. Success probability by required-skill pattern under four models



Each group of bars is P(correct) for examinees holding exactly that combination of the item's required skills (1 = mastered). G-DINA estimates every pattern freely; the three additive models must fit them with main effects only. Where the additive bars miss the G-DINA bar, the item needs an interaction. That happens clearly for V5 (pattern 11) and V6 (patterns 10 and 11).

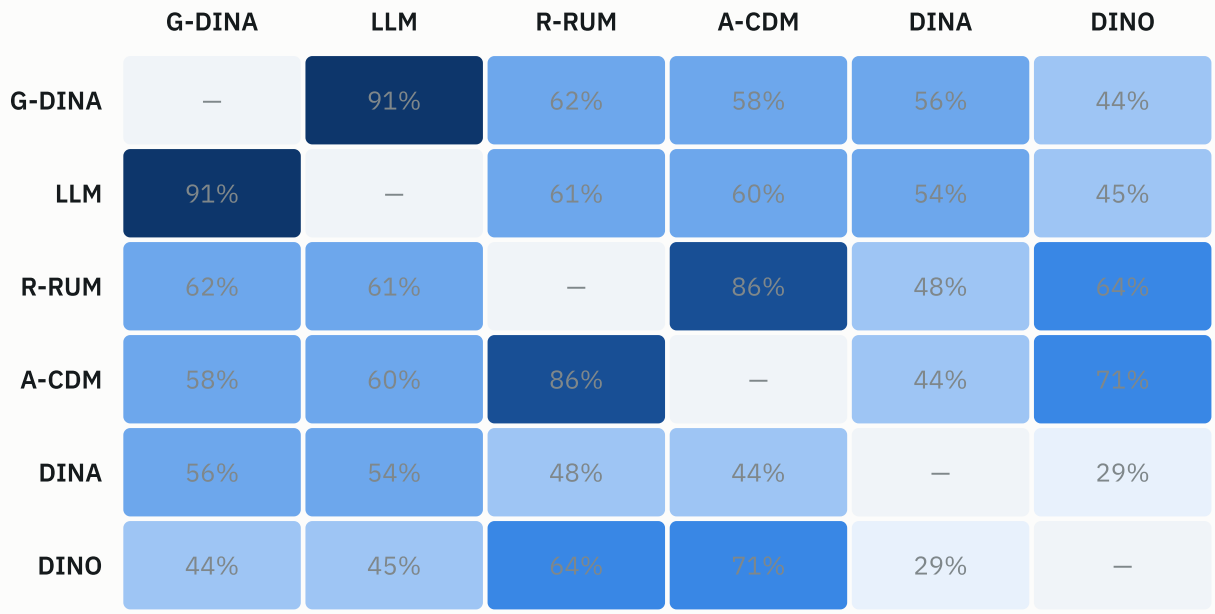
Item	Skills	Rule implied by G-DINA	Guess	Slip	IDI	Monoto
V1	A1	—	0.158	0.172	0.669	
V2	A2	—	0.121	0.234	0.645	
V3	A3	—	0.111	0.090	0.799	
V4	A1 + A3	Additive: each skill helps, both best	0.132	0.163	0.705	
V5	A2 + A3	Conjunctive (DINA-like): needs both	0.114	0.202	0.684	
V6	A1 + A2	Disjunctive (DINO-like): either skill suffices	0.174	0.093	0.733	✕ G-DINA,
V7	A3	—	0.260	0.409	0.331	
V8	A1	—	0.130	0.436	0.435	
V9	A1 + A2 + A3	A2 and A3 drive success; A1 adds nothing	0.138	0.187	0.674	✕ G-DINA, A-CI
V10	A1 + A3	Additive: each skill helps, both best	0.172	0.299	0.530	

Guess, slip and IDI are from G-DINA. The last column lists every model in which PACER flagged a monotonicity violation for the item (DINA and DINO cannot violate monotonicity by construction).

- **V4 and V10 are additive.** Each skill helps on its own and both together help most. G-DINA's interaction term is small (0.22 and 0.01 on the probability scale), so A-CDM, LLM and R-RUM all reproduce them well.
- **V5 is conjunctive.** One skill alone gives no benefit (0.07–0.11), both give 0.80. The additive models can't represent this: A-CDM predicts 0.29–0.34 for one skill and only 0.59 for both.
- **V6 is disjunctive.** Either skill alone gives about 0.9, and the combination adds nothing. A-CDM and R-RUM fit this by making A1 worthless for V6 (0.53 → 0.49), which is one source of their A1 problem. The LLM handles it best of the additive models.
- **V9 points to a Q-matrix error.** Adding A1 lowers P(correct) in G-DINA (000 → 100: 0.14 → 0.05; 010 → 110: 0.53 → 0.31) and has a negative main effect in all three additive models. It is flagged non-monotone in G-DINA, A-CDM, LLM and R-RUM. The data suggest that **V9 does not require A1**.
- **V5 and V6 minor flags.** The G-DINA drops for V5 (0.114 → 0.065) and V6 (0.933 → 0.907) are a few percentage points in small cells and consistent with their AND / OR rules. I don't think they need action.

How much do the models agree about examinees?

Figure 7. Share of examinees assigned the same full skill profile by each pair of models (MAP)



G-DINA and LLM agree on 91% of examinees, and A-CDM and R-RUM on 86%. Across those two families, agreement is only 58–62%. DINA and DINO disagree with almost everything. The models form two camps, and the split between them is almost entirely about A1.

The A1 problem. A1 has two single-skill items, V1 and V8, and they correlate only .28 with each other. Each model family resolves that tension differently:

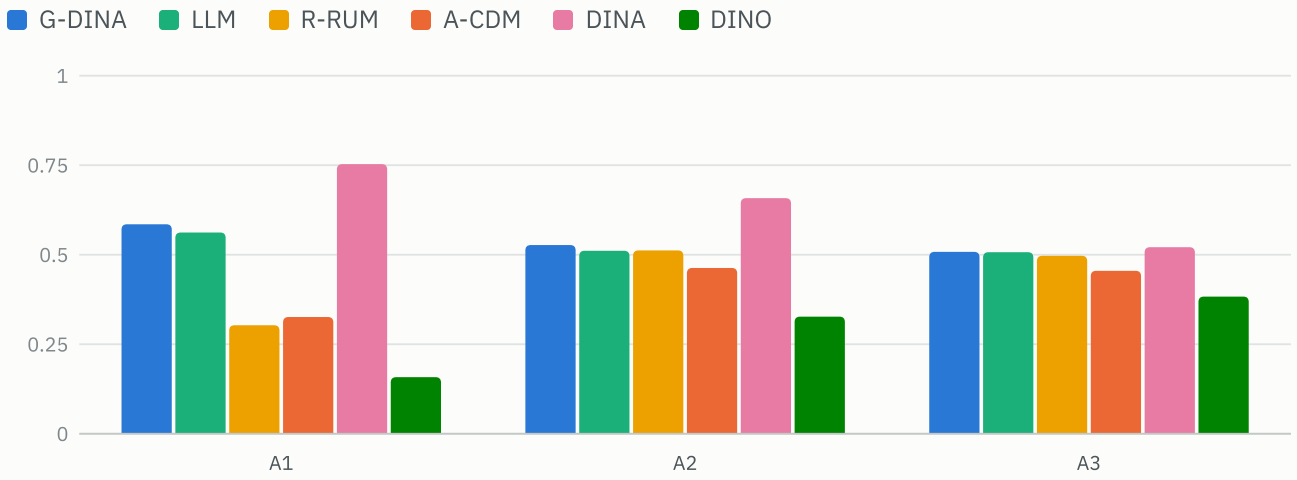
Who counts as an A1 master	A1 mastery rate	P(V1 correct): masters / non-masters	P(V8 correct): masters
G-DINA (MAP)	58%	.88 / .09	
R-RUM (MAP)	32%	.79 / .44	

Under G-DINA, A1 is essentially "can do V1," and V8 is a hard, noisy A1 item. Under R-RUM, A1 is essentially "can do V8": every examinee it classifies as an A1 master answered V8 correctly, and V1 barely separates masters from non-masters. The two models agree on A1 for only 71% of examinees, and 275 examinees who are A1 masters under G-DINA are non-masters under R-RUM. Statistics alone can't say which definition is right. It is a content question about what A1 is and whether V1 and V8 both measure it.

5 Skill mastery profiles

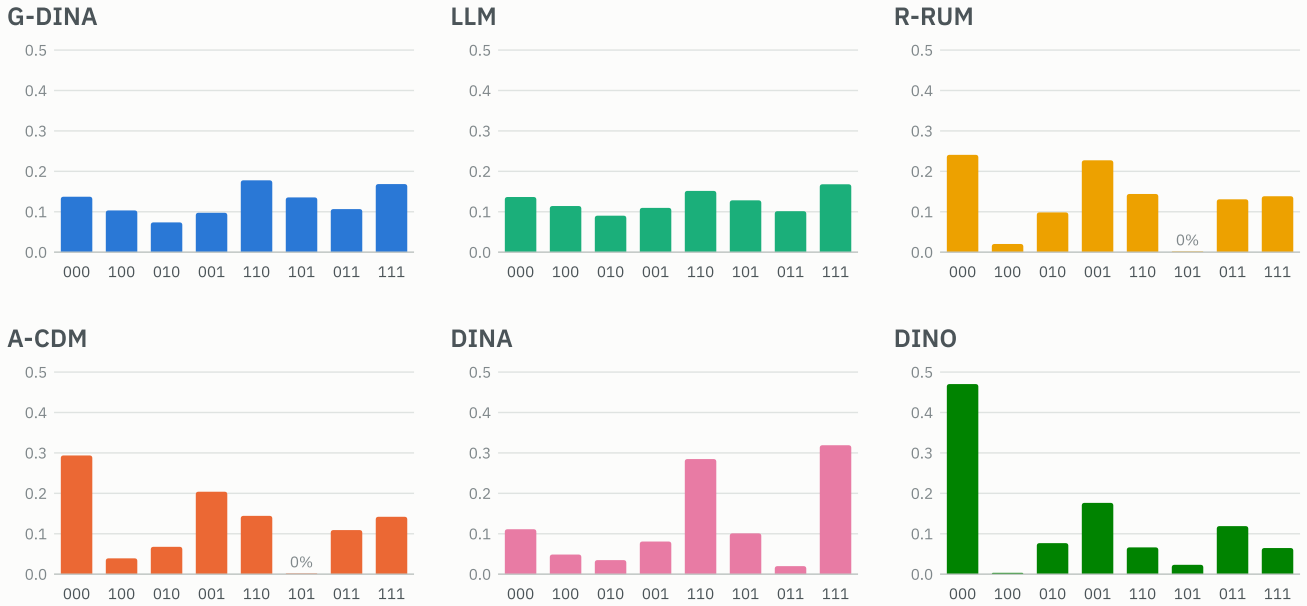
I scored examinees against every CDM calibration with PACER's Skill Profiler (MAP: the most probable profile). For G-DINA I also used EAP (attribute mastered when its posterior probability exceeds .5). G-DINA's EAP and MAP agree for 97.5% of examinees. The profile figures and tables for G-DINA use EAP.

Figure 8. Population skill mastery under each CDM



Estimated population proportion mastering each skill. A2 and A3 are fairly stable across the four non-DINA/DINO models (0.46–0.53). A1 is not: 0.59 and 0.56 under G-DINA and LLM, 0.30 and 0.33 under R-RUM and A-CDM. DINA's "all skills required" rule inflates mastery and DINO's "any skill" rule deflates it.

Figure 9. Estimated distribution of skill profiles (A1 A2 A3) under each CDM



Under G-DINA and LLM, all eight profiles hold 7–18% of examinees. No profile is empty, so there is **no evidence of a strict learning hierarchy**. For example, many examinees master A3 without A1 (001, 011). R-RUM and A-CDM empty the 101 profile and move those examinees mainly into 000 and 001. DINA concentrates 60% of examinees in 110 and 111, and DINO puts 47% in 000. Those patterns come from the models' restrictions, not from the data.

Profile	Skills mastered	G-DINA EAP count	Share	Mean θ (2PL)	SD θ	θ range
000	0	142	14.2%	-1.17	0.36	-1.65 to -0.20
100	1	114	11.4%	-0.54	0.35	-0.99 to 0.38
010	1	71	7.1%	-0.49	0.31	-0.94 to 0.17
001	1	95	9.5%	-0.26	0.31	-0.73 to 0.79
110	2	160	16.0%	-0.18	0.39	-1.16 to 0.64
101	2	132	13.2%	0.35	0.35	-0.35 to 1.02
011	2	110	11.0%	0.63	0.37	-0.30 to 1.36
111	3	176	17.6%	1.13	0.45	-0.14 to 1.81

How certain are the classifications?

Model	Accuracy A1	A2	A3	Mean profile posterior	Profile agrees w/ G-DINA	A1 / A2 / A3 agree
G-DINA	0.88	0.88	0.93	0.74	—	—
DINA	0.88	0.88	0.93	0.76	56%	78% / 80% / 95%
DINO	0.93	0.87	0.90	0.75	44%	61% / 79% / 87%
A-CDM	0.92	0.88	0.92	0.76	58%	71% / 85% / 95%
LLM	0.87	0.87	0.92	0.70	91%	97% / 95% / 98%
R-RUM	0.93	0.89	0.93	0.79	62%	71% / 90% / 98%

Expected accuracy is the average of max(p, 1 – p) over examinees, where p is the posterior probability of mastery. It measures each model's confidence in its own classifications, not agreement with the truth.

Under G-DINA, expected attribute-level accuracy is **0.88 for A1, 0.88 for A2, 0.93 for A3**. That is good for a 10-item test. Classifying the *whole* three-skill profile is harder. The mean posterior probability of the assigned profile is 0.74, and only 42% of examinees are classified with a posterior above 0.80. Report attribute-level results with more confidence than full profiles.

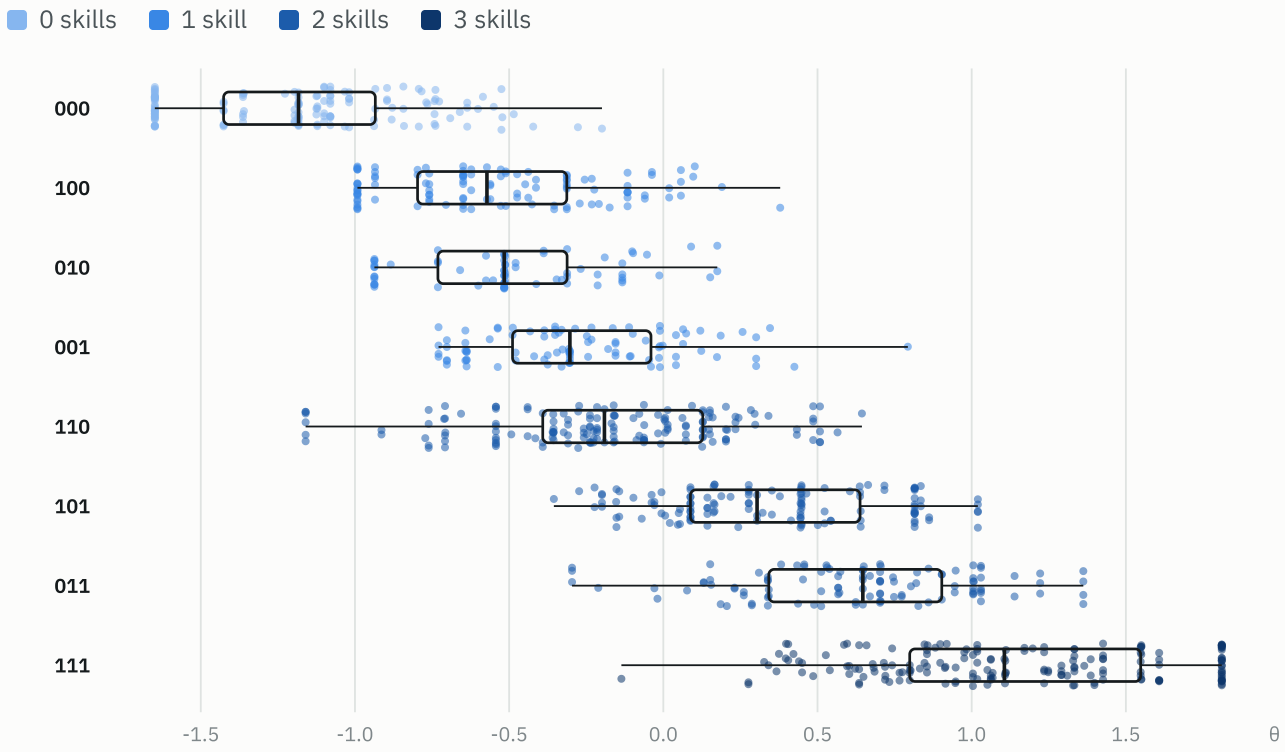
Every model is fairly confident in its own classifications (accuracy 0.87–0.93), yet they disagree with each other. A model's confidence is not evidence that it is the right model. Agreement with G-DINA is 91% for the LLM and 44–62% for the rest. A3 is classified consistently by every model (87–98% agreement), A2 mostly so, and A1 least.

6 Comparing IRT ability with skill profiles

I linked each examinee's 2PL EAP θ to their G-DINA profile. (I verified the linkage: raw scores match row by row in both PACER output files.)

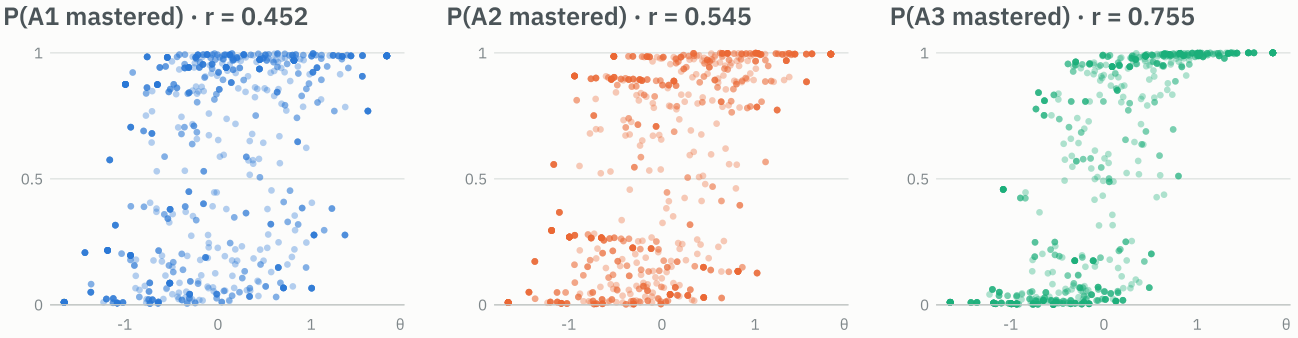
.918	.846	.797	7
r(θ, sum of posterior mastery probabilities)	r(θ, number of skills mastered)	η²: share of θ variance explained by profile	distinct profiles among examinees with raw score 5

Figure 10. 2PL θ by G-DINA skill profile



Each dot is an examinee, jittered vertically. Boxes show the interquartile range and median. θ rises with the number of skills mastered, but the ranges overlap heavily. Within two-skill profiles, mean θ ranges from −0.18 (110) to +0.63 (011). A 110 examinee has about the same θ as a 001 examinee, even though the 110 examinee has mastered two skills and the 001 examinee one.

Figure 11. θ against the posterior probability of mastering each attribute (G-DINA)

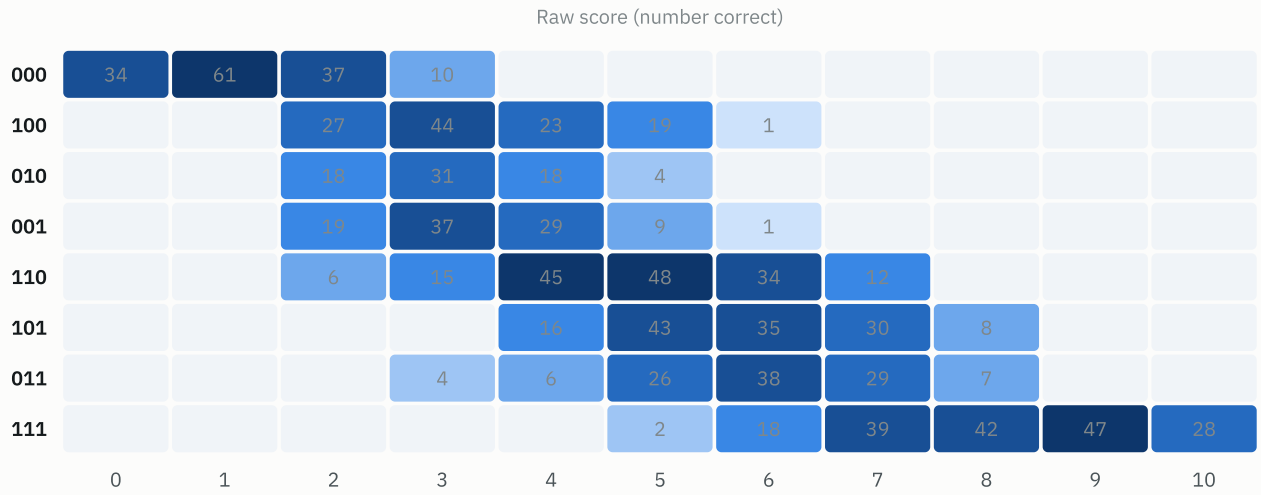


θ tracks A3 most closely (r = .755), then A2 (.545), then A1 (.452). The 2PL θ is effectively an A3-weighted composite. A3 appears on six items, including the most discriminating ones (V3, V5, V9, V10), while A1's single-skill items (V1, V8) have the flattest 2PL slopes.

Model	r(θ, P(A1))	r(θ, P(A2))	r(θ, P(A3))
G-DINA	0.452	0.545	0.755
DINA	0.637	0.528	0.734
DINO	0.541	0.700	0.815
A-CDM	0.462	0.690	0.777
LLM	0.497	0.575	0.758
R-RUM	0.446	0.626	0.746

The pattern holds under every CDM: θ always correlates most with A3 (.73–.82). The A1 correlation stays low (.45–.50) under G-DINA, LLM, A-CDM and R-RUM. It is higher under DINA (.64) and DINO (.54), whose all-or-any rules tie A1 to the multi-skill items that drive θ.

Figure 12. Raw score × skill profile (count of examinees, G-DINA)



Scores of 0–1 map to 000, and scores of 9–10 map to 111. In the middle of the scale, a single raw score (and therefore a single θ) covers four to seven different skill profiles. That middle band is where the diagnostic model adds information θ cannot give.

Interpretation. The two approaches agree on the broad ordering: examinees with more skills have higher θ. They diverge in a way that matters for use. θ can't tell *which* skills an examinee lacks. Among the 151 examinees with raw score 5 (θ ≈ 0.07), 48 are 110 (missing A3), 43 are 101 (missing A2), 26 are 011 (missing A1), and 32 have only one skill. Those groups need different instruction, and the single θ value gives them all the same number. Together with the clear fit advantage of the diagnostic models, this supports treating the instrument as a **diagnostic test of three non-hierarchical skills**, with θ (or the raw score) at most as a secondary overall summary.

7 Items flagged for review

I ranked items by how much evidence across all three frameworks points to a problem.

V7

✕ Priority

2PL worst misfit (Q1 z 10.4) · Lowest IDI in all 6 CDMs (.32–.35) · Slip .35–.42

This is the weakest item under every model. Its empirical curve is non-monotone in θ (Figure 3). In G-DINA, examinees who have mastered A3 answer it correctly only 59% of the time, and non-masters 26%, and every other CDM gives nearly the same numbers. CTT discrimination is acceptable (.34), which hides the problem.

Action: review content and the key. Check for an alternate correct answer or a construct-irrelevant demand. Consider replacing it with a cleaner A3 item.

V9

✕ Priority

Q-matrix suspect · Non-monotone in 4 CDMs · 2PL misfit (4.41)

Adding A1 *lowers* success probability in G-DINA (000 → 100: .14 → .05; 010 → 110: .53 → .31), and A1 has a negative main effect in A-CDM, LLM and R-RUM. PACER flags the item as non-monotone in all four. A2 + A3 alone give .78 in G-DINA. The item otherwise discriminates well (IDI .60–.70).

Action: ask content experts whether A1 is really required. If not, recode V9 as A2 + A3 and refit.

V8

✕ Priority

Pt-bis .293 (marginal) · 2PL a = .39, misfit · Defines A1 in R-RUM/A-CDM · r(V1, V8) = .28

Its meaning depends on the model. Under G-DINA and LLM it is a hard, noisy A1 item (slip ≈ .43, IDI ≈ .44). Under R-RUM and A-CDM it becomes the best item on the test (IDI .83–.85), because those models redefine A1 as "can do V8." Every R-RUM A1 master answered it correctly. V8 and V1, the two A1-only items, correlate only .28.

Action: content review of V8 and V1 together. Decide which one represents A1 as intended, and check V8 for unintended difficulty or a second skill the Q-matrix doesn't record.

V1

▲ Review

Pt-bis .229 (low) · 2PL a = .30, misfit · G-DINA IDI .67 · R-RUM IDI .35

CTT and IRT flag it. Under G-DINA and LLM it is a **good** A1 indicator (masters 83%, non-masters 16–19%). Under R-RUM and A-CDM it is weak (non-masters succeed 43–45% of the time), because those models define A1 by V8 instead. Its weak correlation with the total reflects A1 being weakly related to the rest of the test.

Action: keep it, and review it alongside V8 (see above).

V2

▲ Review

Pt-bis .242 (low) | 2PL a = .39 | IDI .56–.66 across CDMs

The same CTT pattern as V1, for A2, but it is stable across models. It is also the *only* single-skill A2 item, so it is essential for identifying A2.

Action: keep it. Consider adding companion A2-only items.

V5

▲ Review

Hardest item (p = .28) | Conjunctive (AND) | Minor G-DINA monotonicity flag

P(correct) is .80 with both A2 and A3, and at most .11 otherwise. The additive models can't represent this and underpredict the both-skills pattern (A-CDM .59). The small G-DINA dip for one-skill patterns (.065 vs. .114) comes from sparse cells. It is the strongest 2PL discriminator (a = .91).

Action: none required. Note that it behaves as an "AND" item.

V6

▲ Review

Easiest item (p = .73) | Disjunctive (OR) | Non-monotone in 3 CDMs

Either A1 or A2 gives about .9 in G-DINA, and the combination adds nothing (.907 vs. .933). The additive models struggle with it: A-CDM and R-RUM make A1 worthless for V6, and they are flagged non-monotone. That explains its high p-value. The item is diagnostic only at the low end, for examinees who have neither skill.

Action: confirm with content experts that either skill alone should solve it. If so, the Q-matrix is right and the "OR" behavior is intended.

V3

✓ Keep

2PL misfit (6.85) | Most diagnostic item (IDI ≥ .71 in every CDM)

It misfits the 2PL only because it measures A3 very sharply. That also inflates its 3PL slope (2.73).

Action: none.

No items were flagged in any framework for V4 or V10. V3 misfits the 2PL but is the most diagnostic item in the set (G-DINA IDI = .80, and ≥ .71 in every CDM), so I wouldn't change it. Its 2PL misfit is the multidimensionality problem again.

8 Recommendations

1. **Adopt G-DINA as the reporting model** and report attribute-level mastery with posterior probabilities. Reserve full-profile labels for examinees whose profile posterior exceeds about 0.80, or flag the rest as uncertain. If a more parsimonious model is required, use the LLM, not R-RUM or A-CDM.
2. **Resolve what A1 means before reporting A1 scores.** Ask content experts whether V1 and V8 both measure A1. Until that's settled, A1 mastery rates are model-dependent (30% to 59%) and should be reported with that caveat.
3. **Ask content experts to review V9's Q-matrix row**, specifically whether A1 is required. Then refit G-DINA and the LLM with the revised Q-matrix and compare AIC/BIC.
4. **Revise or replace V7.** It has a high slip rate (.35–.42) and the lowest IDI (.32–.35) in every one of the six CDMs.
5. **Strengthen A1 and A2 coverage.** A1 needs a second, cleaner single-skill item that agrees with V1 or V8. Only four items measure A2, and only V2 measures it alone. One or two more single-skill items for each would stabilize classification across models.
6. **If one overall score is needed, use the 2PL θ (or the raw score) with caution.** Reliability is about 0.67–0.68 and SE ≥ 0.6, and θ under-weights A1. Don't use θ cut scores to infer specific skill mastery.
7. **Next analyses outside PACER's current toolset:** item-level model selection (a different CDM rule per item, e.g., DINA for V5, DINO for V6, A-CDM for V4 and V10, chosen by Wald tests), absolute CDM fit (M2/SRMSR), CDM item fit, and Q-matrix validation (e.g., the de la Torre–Chiu PVAF method).

9 Method notes

- **Everything statistical was estimated in PACER 3.2.0**: CTT item analysis (pairwise, adjusted point-biserial); IRT calibration (MML-EM, PACER defaults, 3PL with its default beta prior on *c*); EAP scoring (N(0,1) prior, 49 quadrature points); TIF/SEM curves and item fit plots; DINA, DINO, A-CDM (identity link), LLM (logit link), R-RUM (log link), and G-DINA (identity link) calibration (EM, tolerance 1e-4, 3 starts, seed 123456); and Skill Profiler scoring (MAP for every CDM, plus EAP for G-DINA).
- **Computed by me from PACER output:** AIC/BIC for the IRT models (PACER reported −2LL and parameter counts), likelihood-ratio tests, marginal reliability, the merge of θ with profiles, correlations, η^2, cross-tabulations, classification-accuracy summaries, between-model

agreement rates, and the V1/V8 conditional proportions. All are simple arithmetic on PACER's per-examinee output files and the response file. No models were estimated outside PACER.

- The comparability of IRT and CDM likelihoods assumes both are marginal likelihoods of the same 1,000 × 10 response matrix, which they are. They use different latent distributions (normal vs. 8-class categorical), and the comparison is by information criteria only.
- The file names suggest these may be simulated G-DINA data. If so, the results are consistent with that origin: the saturated model fits best by AIC and the likelihood-ratio test, and items follow different condensation rules. The V9/A1 finding would then point to a weak or near-zero A1 effect in the generating parameters, not to a content error.